# The Value Chain: A Study of the Coffee Industry

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## Introduction

This chapter presents a case study of certain aspects of a transfer pricing study of a tested party roasting operation in Europe that is a related party of a hypothetical global specialty coffee company.[1] The certain aspects include an identification of the supply chain of the coffee specialty industry, a value chain analysis of the industry, and the identification of a pool of potential comparable parties for the tested party.

## The Supply Chain

The supply chain is the set of activities required to transform a raw product to the finished goods sold to consumers. A comprehensive supply chain study begins with a literature review, from bean to cup. Scholarly and industry sources discussing the supply and value chain allowed for the identification of many plausible determinants of value throughout the coffee process. However, there is not literature addressing the transfer pricing aspect of a coffee roasting operation.

For coffee, the supply chain typically includes three main stages (*see* Figure 1):

Stage 1: agricultural steps that includes coffee cultivation, harvesting.
Stage 2: treatment and processing up to the green coffee and production steps consisting of roasting and grinding coffee.
Stage 3: retailing, consumption and disposal steps.

The coffee chain starts with cultivation that includes planting coffee trees, applying fertilizer, pesticides and herbicides, irrigating the plants and finally harvesting coffee cherries. After cultivation, coffee cherries are treated by dry/wet processing to get the green coffee. Then, a variety of measures are applied in the refining process consisting of polishing, sorting, washing and drying, which is followed by roasting process. After roasting and packaging, coffee is ready for retailing and consumption. Consuming coffee requires water and energy, which are the most concerned environmental issues in the coffee life cycle. Finally, coffee grounds, filters and packaging are disposed by consumers. Note that in the coffee supply chain, the transportation distances are the most important concerns of shipping green and roasted coffee from producing to consuming countries.[2]

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[1] The specialty coffee roasting global company for this case study is fictitious. No identification with an actual company is intended or should be inferred.

[2] *Id*.

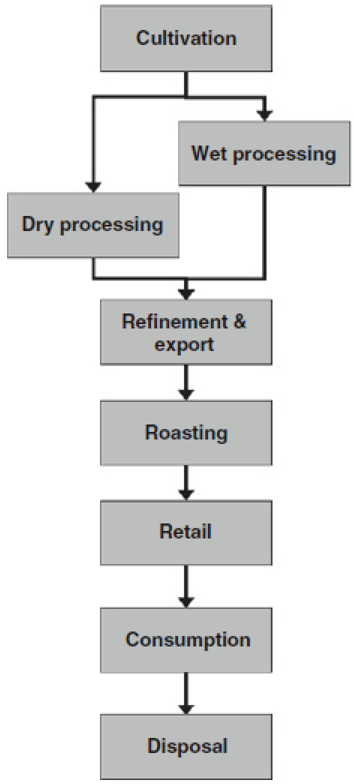


**Figure 1. Coffee Supply Chain**[3]

There are number of players involved in the coffee chain (*see* Figure 2): the local agents who collect cherries from small planters, the wholesalers who function as traders and are exporter/importer of the beans, the roasters, the retailers and the consumers.[4] Among the players in the coffee chain, roasters have the highest influence on the other actors. Particularly, they dominate international traders as a result of oversupply, market concentration, increased flexibility in blending, and adopting "supplier-managed inventory" strategies, which passes risk on to the other participants within the chain.[5] By implementing "supplier-managed inventory," a proportion of the stocks move to traders, thus the roaster holds a minimum quantity linked to projected sales of roasted coffee.[6]

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[3] *See* Viere, T., von Enden, J., & Schaltegger, S. (2011). Life cycle and supply chain information in environmental management accounting: a coffee case study. In Environmental Management Accounting and Supply Chain Management (pp. 23-40). Springer Netherlands.
[4] Depperu, D., & Todisco, A. (2010). Value creation in the fair trade chains (No. 4). Collana Working Paper del Centro di Ricerca per lo Sviluppo Imprenditoriale.
[5] Ponte, S. (2002). The Latte Revolution? Regulation, Markets and Consumption in the Global Coffee Chain. World Development 30 (7), 1099 – 1122.
[6] Daviron, B., & Ponte, S. (2005). The coffee paradox: Global markets, commodity trade and the elusive promise of development. Zed books. Chicago.

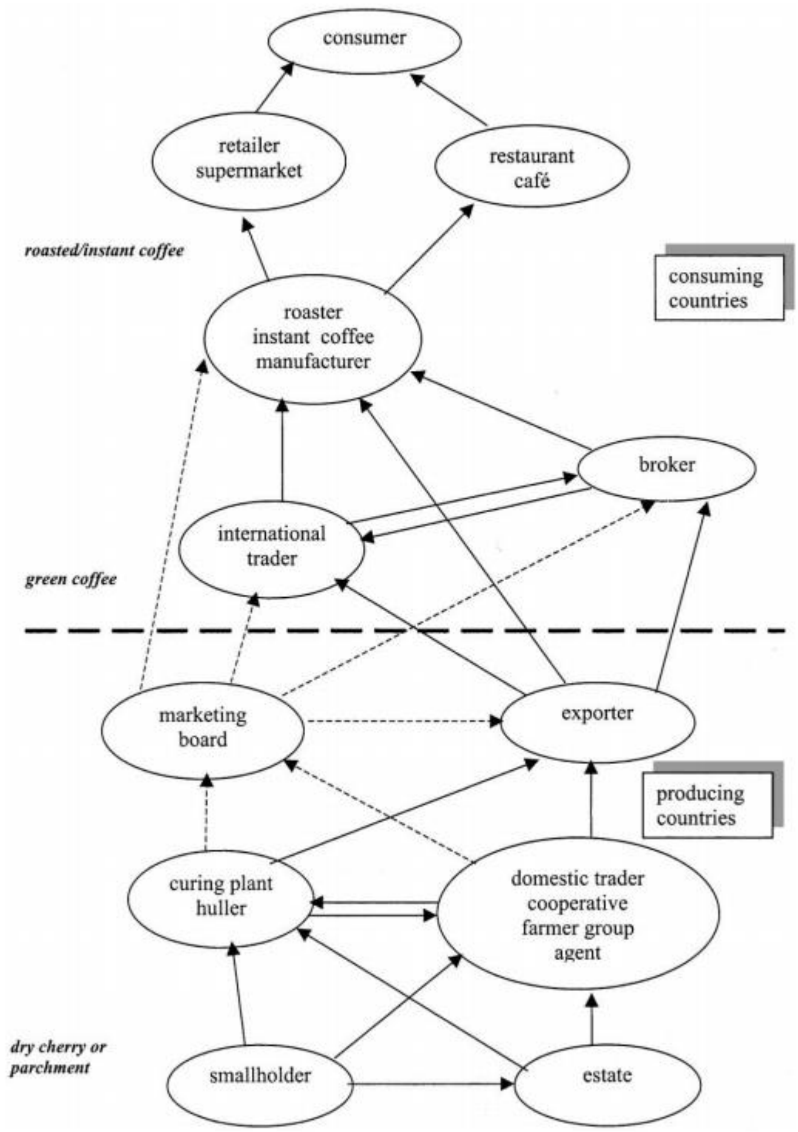


**Figure 2. General Structure of the Coffee Chain**[7]

## The Value Chain

The value chain differs from the supply chain in that the value chain requires allocating value to each stage of the supply chain.[8] This value chain analysis estimates the distribution of total income or surplus along the coffee chain.[9] Surplus is the net retained profit at each of the coffee chain stages. It is possible to estimate the retained values as a percentage of the retail price. Retained value identifies the income that remains within each producing and consuming country. In the consuming country, it is

[7] Ponte, S. (2002). The Latte Revolution? Regulation, Markets and Consumption in the Global Coffee Chain. World Development 30 (7), 1099 – 1122.
[8] Sarris, A. (2008). Commodity Market Review 2007-2008. FAO Trade and Markets Division.
[9] Talbot, J. M. (1997). Where does your coffee dollar go?: The division of income and surplus along the coffee commodity chain. Studies in comparative international development, 32(1), 56-91.

measured by deducting the costs of imported green coffee from the retail price, which is equivalent to the total value added in the consuming country. In the producing countries the total value added is estimated by deducting the grower's price from export unit value.[10]

The total income along the coffee chain yields from the money spent by consumers to purchase coffee products. A study by Professor John M. Talbot divides the total income per pound of coffee into four parts: (1) the total income of coffee growers; (2) the incomes of processors, traders, exporters, and the state within producing countries; (3) the incomes of coffee manufacturers, wholesalers, retailers, and the state in consuming countries; and (4) a residual category including shipping costs, profits of shippers and financiers, and weight losses that occur during the shipping and roasting of the coffee.[11] The share of margin among the coffee chain is allocated in the following order: [12]

| | |
|---|---|
| the producers: | 20 percent |
| local intermediaries: | 11 percent |
| international traders: | 8 percent |
| exporters: | 6 percent |
| insurance and freight fees of importing/exporting: | 2 percent |
| roasters: | 30 percent |
| retailers: | 15 percent |
| taxes: | 6 percent. |

Based on this description, derived from a literature review, of the value chain of the coffee industry, the roasters receive the highest share of the retail price. Roasters are the key actors in the coffee chain, especially as the roasters are able to generate not only the highest profit margin but to capture the most value along the value chain.[13] Roasting is a very capital-intensive operation. One may approach identifying the profitability of different links by assessing the links from a ratio of capital employed by the group rather than the share of total price.[14]

Moreover, the knowledge of roasting cost is essential as a guide in connection with estimating the roaster's profitability. The taste of coffee is determined in the roasting process by blending different kinds of coffee from different places.[15] The most important operation for a roaster is blending, the one in which specific know-how is mobilized.[16] Since major roasters receive their value because of their flavor and as a result of blending, the product's identity is more closely associated with the brand name than with its geographical origin.[17]

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[10] *Id*.
[11] *Id*. Professor John M. Talbot, University of West Indies (Mona).
[12] Depperu, D., & Todisco, A. (2010). Value creation in the fair trade chains (No. 4). Collana Working Paper del Centro di Ricerca per lo Sviluppo Imprenditoriale.
[13] *See* Raynolds, L. T. (2002). Consumer/producer links in fair trade coffee networks. Sociologia ruralis, 42(4), 404-424. If coffee is roasted in the producing country, the bulk of the value would be captured in consuming countries by distributors and the retailers. *See also* Stiller, R. (2015). Small Beans-Big Business: Existing Opportunities for Producing Countries in the International Coffee Trade (Doctoral dissertation).
[14] Fitter, R., & Kaplinsky, R. (2001). Can an agricultural 'commodity' be de-commodified, and if so who is to gain?
[15] Borzoni, M. The Green Coffee Purchasing Policies of Italian Roasters.
[16] Daviron, B., & Ponte, S. (2005). The coffee paradox: Global markets, commodity trade and the elusive promise of development. Zed books.
[17] Renard, M. C. (1999). The interstices of globalization: The example of fair coffee. Sociologia ruralis, 39(4), 484-500.

The largest coffee roasters are Nestlé, Kraft, Sara Lee, Smucker's (P&G), Dalmayr, Starbucks, Tchibo, Aldi, Melitta, Lavazza, and Segafredo.[18] A difference in practice between the largest roasters and smaller roasters is that the largest generally rely upon traders to generate their supply of coffee rather than dealing directly with producers or producer groups, whereas smaller, especially specialty, roasters tend to secure their supplies directly from producers and their organizations.[19] The coffee value chain is shown in Figure 3 (Note that the figures provided in Figure 3 refers to a particular period of time in 1994).

The cost of the roasted coffee incorporates the cost of the green coffee, the expense of roasting it and the shrinkage cost caused by the roasting. The roasting expenses are classified into two main categories: direct labor and overhead. The overhead expense consists of the following:[20] (1) superintendence, (2) power, (3) lighting and heat, (4) fuel and supplies, (5) machinery and equipment repairs and maintenance, (6) depreciation of machinery and equipment, (7) building expense (exclusive of depreciation), (8) depreciation of building, (9) insurance, (10) rent, and (11) sundry roasting expense.

In brief, a share of income along the coffee chain is received by each following categories:[21]

- Farmers that pick and process cherries receive a 'farm gate price'.
- Further processed cherries are traded at a 'factory gate price'.
- The green beans that are available to export by intermediary are sold at free on board prices (fob) prices.
- The shipped beans to consuming countries are exchanged for cost, insurance and freight (cif) prices.
- The beans are passed at wholesale prices.
- Roasters sell the processed beans at factory gate prices.
- Distributers sell the coffee to the end consumers at retail prices.

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[18] *See* Tropical Commodity Coalition for Sustainable Tea Coffee Cocoa (TCC). 2009. Coffee Barometer 2009. TCC: The Hague. Coffee Coalition (CC). 2006. *See also* Coffee Barometer 2006. Certified Coffee in The Netherlands. CC: The Hague. *See also* Quo, S., El Café, D. Y. O. P., & Se, A. Q. Status Quo, Challenges And Opportunities Of Alternative Coffee That Is Produced In México And Consumed In Germany.
[19] Van Dingenen, Katrien & Koyen, Marie-Laure & Koekoek, Freek-Jan & Pierrot, Joost & Giovannucci, Daniele (2010). European and Belgian market for certified coffee. Final report CTB/BTC NoBXL 609 – lot 3.
[20] Kester, R. B. (1918). Accounting theory and practice (Vol. 3). Ronald Press Company.
[21] Fitter, R., N 12, *supra*.

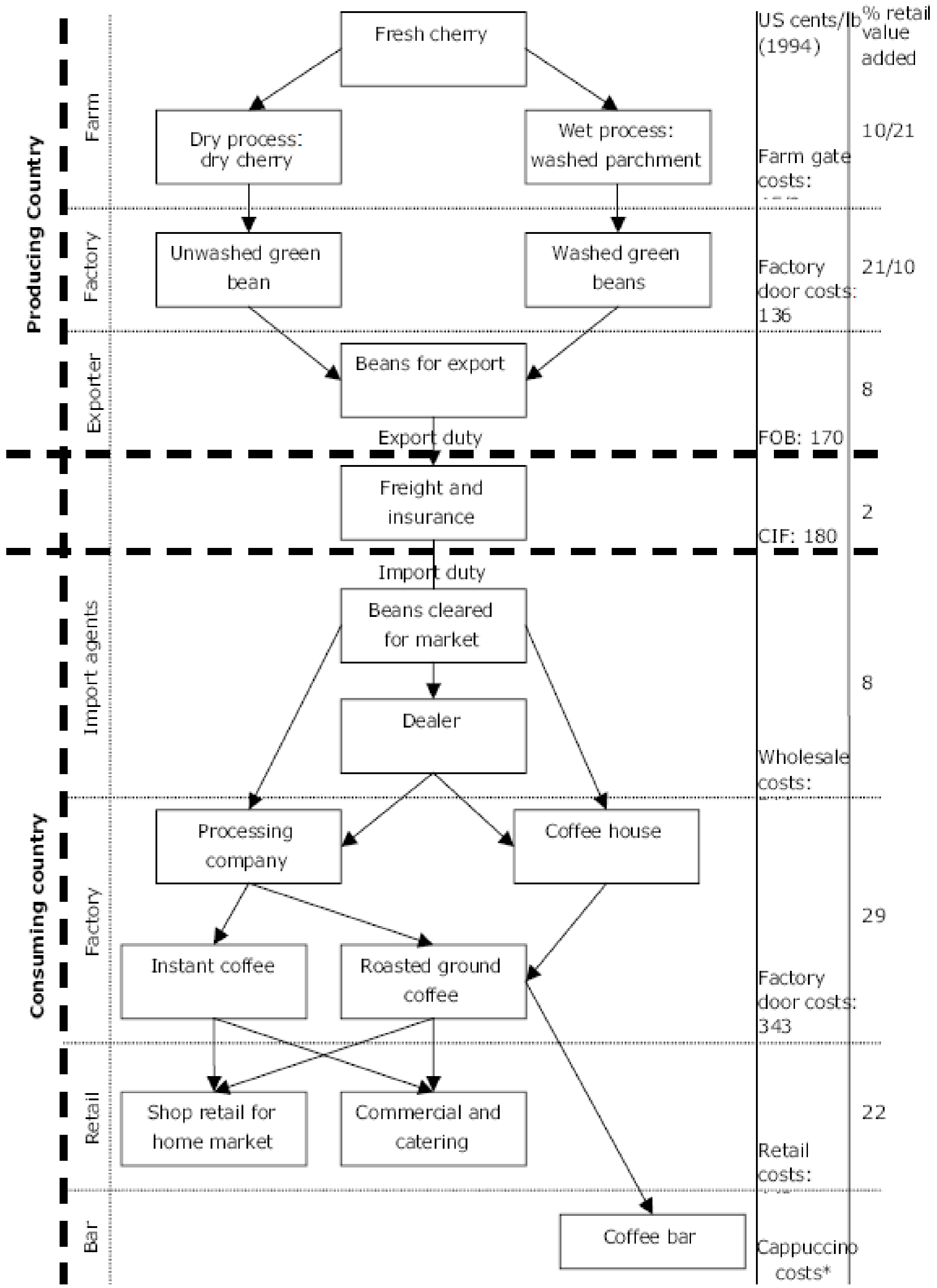


**Figure 3. The Coffee Value Chain[22]**

[22] Fitter, R., & Kaplinsky, R. (2001). Can an agricultural 'commodity' be de-commodified, and if so who is to gain?

# Identifying a Pool of Potential Comparable Parties

Interviews were undertaken with data experts, industry professionals and academics. These individuals assisted with the search, scoping and qualification of the data used for the value chain analysis. Experts include Dr. James Mcclure of Thomson Reuters; Brett Schoell of KTmine; Dr. Ednaldo Silva, Marcia Coutinho and Julia Vasconcellos from RoyaltyStat; and Dr. Lorraine Eden from Texas A&M University.

Databases used:

- Industry wide indicators:
  - BizMiner by Brandow Company, Inc.
- Company Financials:
  - S&P Capital IQ Compustat
  - One Source by Thomson Reuters
  - ORBIS
- Royalties and License Agreements:
  - KtMine
  - RoyaltyStat
- Market Prices:
  - International Coffee Organization

Public and privately held companies from the above listed sources were filtered for developing a pool of potential comparable parties for the hypothetical coffee roaster. In order to retrieve the appropriate financial indicators from the adequate comparables, a search strategy was developed. Dataset were filtered by using the main economic activity of the company. One or more of the following codes was applied in the search strategy:

- NAICS: 311920 (Coffee and Tea Manufacturing)
- SIC: 2095 (Roasted Coffee)
- NACE: 1083 (Processing of tea and coffee)

Along with the industry codes, a text search was undertaken for the word 'coffee' in the description of the companies' activities. The search was limited to a regional location, focusing on companies situated in Europe. The number of coffee roasting companies situated in the country chosen for the study, The Netherlands, did not produce sufficient for statistical analysis. The pool of companies in all of Europe generated a sample of over 7704 companies. Data from public authorities and companies with no recent financial data was excluded. The results were then calibrated downward to a set of 2981 by excluding companies with no operating revenue data available. The rationale was that in the dataset, companies with no reported operating revenue had incomplete data or no financial data at all. The countries with the most and least firms are listed below. A transfer pricing advisor will want to at this stage further scope the pool of potential comparable parties by undertaking the comparables analysis discussed in Chapter 2, *infra*.

**Table 1**

| Country | # of Companies |
|---|---|
| Italy | 720 |
| France | 309 |
| Russia (Europe) | 284 |
| Spain | 195 |
| Romania | 123 |
| ... | … |
| Netherlands | 2 |